

Bioinspired carbon dot-Ag nanocomposite derived from basil leaves for sensitive electrocatalytic detection of ciprofloxacin

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ABSTRACT

Rapid identification of antibiotic residues is crucial for environmental surveillance and ecological balance. In this study, carbon dots (CDs) were synthesized using a green approach with Thai basil leaves and utilized for the synthesis of carbon dot-silver nanoparticles (CD-AgNPs) nanocomposite for electrochemical sensing of ciprofloxacin (CPX) in environmental samples. Morphological characterization revealed predominantly spherical CDs and CD-AgNPs, with an average diameter of 5 nm and 20.24 ± 2.5 nm, respectively. Modification of the glassy carbon electrode (GCE) with CD-AgNPs increased the effective surface area (A_e) from 0.040 to 0.087 cm², and the charge transfer resistance (R_p) decreased significantly from 2842 to 851 k Ω . Surface coverage (Γ^*) of CPX was 6.593×10^{-11} mol cm⁻². Electrochemical studies revealed a two-electron oxidation mechanism with a rate constant (k^0) of 0.1157 s⁻¹ and a standard peak potential (E^0) of 1.077 V, leading to CPX radical generation. The sensor achieved an impressive limit of detection (LOD) (0.1124 μ M), limit of quantification (LOQ) (0.3746 μ M), and sensitivity (0.7977 μ A μ M⁻¹ cm⁻²) within the 0.25–40 μ M range. Selectivity tests indicated minimal interference from other antibiotics, and signal degradation was limited to 11.4% after 10 consecutive measurements. Application to tap and river water demonstrated recovery rates of nearly 100%, aligning with high-performance liquid chromatography (HPLC) analysis. Overall, Thai basil-derived CD-AgNPs demonstrate high-performance electrochemical sensors for efficient CPX detection.

1. Introduction

The latest advancements in electrocatalytic detection systems have markedly enhanced the detection and quantification of therapeutic compounds, for instance, CPX. CPX, a widely used fluoroquinolone-class drug, is a prominent agent in the treatment of bacterial infections because of its strong antimicrobial activity and excellent tissue penetration. However, due to its extensive clinical use and improper disposal, considerable amounts of CPX are continuously introduced into aquatic environments through domestic wastewater, hospital effluents, and pharmaceutical discharge [1]. Its persistence in natural water bodies has raised serious environmental concerns, as CPX residues can disrupt aquatic ecosystems, alter microbial communities, and promote the proliferation of antibiotic-resistant bacteria [2]. Consequently, the

environmental occurrence of CPX is increasingly recognized as a critical ecological and public health issue [3]. Therefore, sensitive and reliable detection of CPX in environmental samples is essential for ongoing surveillance and risk assessment.

Traditional analytical techniques such as chromatography—particularly HPLC and liquid chromatography–mass spectrometry, as well as chemiluminescence, capillary zone electrophoresis, surface-enhanced Raman scattering [4], UV–visible, and fluorescence spectroscopy, have proven highly selective, sufficiently sensitive and precise for detecting and quantifying pharmaceutical contaminants in environmental samples [5–9]. Despite their reliability, these methods are generally labour-intensive, expensive, and involve complex sample preparation steps, along with the need for trained personnel and costly reagents. As a result, growing attention has shifted toward the development of simpler,

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low-cost electrochemical sensing systems employing plant-based nanomaterials [10,11]. Electrochemical sensing systems have attracted considerable interest for their effectiveness in the rapid and precise detection of organic contaminants, including pesticides and pharmaceutical compounds [12]. This approach is appealing due to its straightforward electrode modification process, cost-effectiveness, and ability to deliver high sensitivity and selectivity, even at low analyte concentrations. It also supports rapid, in-field environmental assessments. Electrochemical tools, including cyclic voltammetry (CV), chronoamperometry, differential pulse voltammetry, and square wave voltammetry (SWV), are often employed to probe the electrocatalytic performance of nanomaterial-modified electrodes toward pharmaceutical compounds. Among these, SWV provides the most distinct analytical advantages, particularly in the precise detection of trace quantities [8,13].

In recent years, NPs and nanocomposites have been broadly looked into for modifying various electrode materials, namely glassy carbon, platinum, gold, carbon paste [14], and sonogel-carbon [15]; among these, GCE offer distinct advantages over others owing to their excellent mechanical stability, wide electrochemical potential window, and high chemical inertness, enabling low background currents and highly reproducible, durable performance in analytical applications. These nanomaterials are often dispersed in polymeric binders like Nafion, polyvinyl chloride, polystyrene, polyethylene glycol, silicone, or polytetrafluoroethylene to ensure uniform coating and adhesion [11,16]. Incorporation of NPs and nanocomposites enhances the electrochemical behavior of sensors by providing a larger active surface area, superior electrical conductivity, and improved catalytic activity [17]. Among the available binders, Nafion D-521 was selected for its excellent film-forming and adhesive properties, chemical stability, and charge transport, as further supported by the excellent repeatability and stability results demonstrating consistent current response and durable immobilization of the modified electrode [18].

Among the different electrode materials that have been explored, carbon-based nanostructures—especially CDs obtained through eco-friendly or biomass-derived routes—have attracted growing scientific interest. This is due to their zero-dimensional structure, which has a particle size under 10 nm, being hydrophilic, exhibiting good biocompatibility, and possessing outstanding photoluminescence characteristics. The various properties of CDs, like high surface area, plentiful oxygenated surface groups such as carboxyl, epoxy, carbonyl, and hydroxyl, excellent electrical conductivity, and high sorption capacity, were found to facilitate faster electron transfer processes, lower overpotentials, and enhanced electrochemical responsiveness. CDs have demonstrated significant potential in electrochemical monitoring of pharmaceutical analytes such as chlorpyrifos, and azathioprine [10,19]. An earlier study reported a CD-containing biohybrid system for environmentally relevant pollutant treatment, highlighting the growing role of CD hybrids in water-quality management [20]. AgNPs represent another versatile material commonly applied in medical, environmental, and industrial sectors [21,22]. Owing to their exceptional physicochemical properties, namely an enhanced surface-to-volume ratio, efficient charge transport, superior stability, top conductivity, and excellent catalytic efficiency, AgNPs have found increasing use in recent years, particularly for the preparation of functionalized electrodes. It has found extensive applications across multiple fields, including electrochemical and optical sensors [11,21]. In this context, nanocomposite-based electrodes have been increasingly explored for the electrochemical sensing of CPX. For instance, Liang and colleagues reported the detection of CPX by reduced graphene oxide-loaded spherical AgNPs with a LOD of 6.25 nM [23]. Mekgoe et al. [24] reported the synergistic electrochemical detection of CPX using a bismuth vanadate nanocomposite-modified activated carbon electrode derived from banana peel biomass with a LOD of 0.1279 μ M. Patel and colleagues reported one-pot synthesis of β -cyclodextrin modified AgNPs for highly sensitive detection of CPX with a 0.028 nM LOD [25]. Recent interfacial

engineering of hybrid nanocomposites, such as the $\text{NH}_2\text{-MIL-125(Ti)}/\text{Ti}_3\text{C}_2\text{T}_x$ heterojunction [26], highlights how rational composite design enhances electrochemical sensitivity for trace organophosphate detection. Among the various synthesis methods of AgNPs and related nanocomposites, the green synthesis route is eco-friendly, cost-effective, and minimizes the use of environmentally intensive chemicals. In line with this sustainable approach, the present study reports, to the best of our knowledge, one of the few examples of a Thai basil leaf-derived CD–AgNPs nanocomposite synthesized via a green method, along with its systematic structural characterization and application as an electrocatalyst for the electrochemical detection of CPX in environmental water samples.

This study introduces a simple, sustainable, and cost-efficient technique for generating a CD–AgNPs nanocomposite, where CDs function as both reducing and stabilizing agents. *Ocimum basilicum* var. *thyriflora*, commonly known as Thai basil, is an aromatic culinary and medicinal plant widely cultivated across Southeast Asia. It contains a rich profile of bioactive phytochemicals, including essential oils, polyphenols, flavonoids, and phenolic acids, which are characterized by diverse carbon, nitrogen, and oxygen containing functional groups [27]. These inherent chemical moieties can be directly utilized in the formation of CDs, removing the need for additional passivation or doping agents. Therefore, Thai basil represents a sustainable and bioactive source for synthesizing CDs via a convenient single-step hydrothermal procedure. The CDs, obtained from Thai basil leaves through hydrothermal treatment, are zero-dimensional carbon nanostructures under 10 nm in size, capable of facilitating controlled reduction and stabilization of metal nanoparticles for nanocomposite fabrication. Thereafter, the CD–AgNPs nanocomposite was immobilized using a binder on the surface of bare GCE for the electrocatalytic detection of CPX by the SWV technique in environmental samples. The practical applicability of the developed electrochemical sensing system was evaluated in environmental samples, including tap and river water, using SWV and benchmarked against HPLC analysis.

2. Materials and methods

2.1. Chemicals and materials

The study used analytical-grade reagents, and utilized without any further processing unless otherwise specified. Silver nitrate (AgNO_3 , $\geq 99\%$), sodium chloride (NaCl , $\geq 99\%$), hexammine ruthenium(III) chloride ($\text{Ru}(\text{NH}_3)_6\text{Cl}_2$, $\geq 98\%$), potassium chloride (KCl , $\geq 99\%$), disodium hydrogen phosphate (Na_2HPO_4 , $\geq 99\%$), sodium dihydrogen phosphate (NaH_2PO_4 , $\geq 99.25\%$), sodium hydroxide (NaOH , $\geq 98\%$), hydrochloric acid (HCl , 36–38%), ethanol ($\text{C}_2\text{H}_6\text{O}$, $\geq 99\%$), and chloramphenicol ($\text{C}_{17}\text{H}_{12}\text{Cl}_2\text{N}_2\text{O}_5$, $\geq 99.78\%$) were obtained from Merck, India. CPX ($\text{C}_{17}\text{H}_{18}\text{FN}_3\text{O}_3$, $\geq 98\%$), chloroquine ($\text{C}_{18}\text{H}_{26}\text{ClN}_3 \cdot 2\text{H}_3\text{PO}_4$, $\geq 98\%$), phenylbutazone ($\text{C}_{19}\text{H}_{20}\text{N}_2\text{O}_2$, $\geq 98\%$), amoxicillin ($\text{C}_{16}\text{H}_{19}\text{N}_3\text{O}_5\text{S}$, $\geq 98\%$), and sulfamethoxazole ($\text{C}_{10}\text{H}_{11}\text{N}_3\text{O}_3\text{S}$, $\geq 98\%$) were purchased from Sigma-Aldrich, India. Polyvinylpyrrolidone (PVP K-30, $(\text{C}_6\text{H}_9\text{NO})_x$) were procured from SRL (India). A 5% (w/w) Nafion D-521 solution in water/1-propanol was obtained from Alfa Aesar (USA). A CPX stock solution was formulated in ethanol (highly pure), whereas all other solutions for experimentation were prepared using deionized (DI) water from Millipore system (Elix-3, USA).

2.2. Synthesis of CDs

Fresh Thai basil leaves were collected from the local market near Chulalongkorn University, Bangkok, Thailand. Collected fresh basil leaves were washed thoroughly with DI water followed by Milli-Q water, before use to remove the dust, debris, or other unwanted contaminants present on the leaves surface. The leaves were allowed to dry (37°C) in a heating oven to remove all the moisture content and then crushed into fine powder to lower the impact of seasonal fluctuations on the chemical

composition. The dried powder was utilized for all subsequent experiments. CDs were synthesized via a hydrothermal approach following a relevant prior art, and the optimized reaction conditions established therein were adopted in the present study [28]. Specifically, 1 g of powder was added to 50 mL of Milli-Q water and stirred for 15 min to make a homogeneous blend. Following transfer into a 100 mL Teflon-lined autoclave, the mixture underwent hydrothermal treatment at 200 °C for 10 h. After reaching room temperature, the yellow suspension was subjected to centrifugation for 20 min at 14,000 rpm and subsequently passed through a 0.22 μm membrane filter to remove coarse particles. The final product was refrigerated at 4 °C for subsequent experiments. However, in this study, CDs serve solely as a green reducing/stabilizing agent for CD-AgNPs formation rather than as the primary sensing probe. Therefore, reproducibility was evaluated at the functional level of the CD-AgNPs nanocomposite. Independently prepared CD batches consistently produced CD-AgNPs with comparable morphology, charge-transfer resistance, and electrochemical sensing performance (LOD, sensitivity, recovery). Hence, additional statistical yield-based batch acceptance criteria are not expected to influence the conclusions of this study.

2.3. Synthesis of CD-AgNPs nanocomposite

The CD-AgNPs nanocomposite was synthesized using a green and facile method following a previously reported and optimized protocol [29]. Briefly, 0.5 mL of CDs solution was added to 10 mL of freshly prepared 5 mM AgNO_3 solution, and the mixture was stirred vigorously for 1 min, followed by incubation at room temperature for 2 h. The reduction of Ag^+ ions by the CDs led to the formation of a reddish-brown CD-AgNP dispersion, as illustrated in Fig. 1. No external pH or ionic strength adjustment was applied, as the reaction proceeded under the intrinsic chemical environment provided by the CD solution, consistent with the optimized protocol. The synthesized composite was collected by centrifugation (14,000 rpm, 10 min), washed three times with DI to remove unreacted species, and stored at 4 °C for subsequent applications.

2.4. Preparation of CD-AgNPs/GCE

A 3 mm diameter GCE was meticulously polished using alumina slurry (0.3 μm) and cleansed completely with DI water, and dried at 25 °C. To prepare the CD-AgNPs modified GCE (CD-AgNPs/GCE), CD-AgNPs nanocomposite was dispersed in an equal volume (100 μL) of Nafion solution (0.5%) in isopropanol and subjected to ultrasonication (45 min) to ensure uniform dispersion. Subsequently, 5 μL of the resulting CD-AgNPs solution was drop-cast onto the cleaned GCE surface and dried at 25 °C prior to electrochemical characterization.

2.5. Characterization of CDs and CD-AgNPs

The optical and structural features of the synthesized CDs and CD-AgNPs were assessed using multiple analytical instruments. To validate the successful synthesis, ultraviolet-visible (UV-Vis) spectra were recorded in the 200–700 nm region with a Shimadzu double-beam spectrophotometer (Japan). Fluorescence emission spectra were recorded using a PerkinElmer LS 45 spectrofluorometer (Singapore). The morphological and structural characteristics of the synthesized samples were analyzed using field emission transmission electron microscopy (FETEM) with a JEM-2100 instrument (JEOL, Japan). X-ray diffraction (XRD) studies were carried out using a SmartLab 9 kW diffractometer (Rigaku, Japan) with $\text{Cu-K}\alpha$ radiation ($\lambda = 0.15406$ nm), scanning from $2\theta = 10^\circ$ – 80° at a rate of $20^\circ \text{ min}^{-1}$ to investigate the crystalline nature of CDs and CD-AgNPs. Fourier-transform infrared (FTIR) spectroscopy (IRAffinity-1, Shimadzu, Japan) was employed to determine the functional groups in the synthesized samples in the range of 4000 – 500 cm^{-1} . Additionally, elemental composition and surface chemistry were inspected via X-ray photoelectron spectroscopy (XPS) employing the Thermo Fisher Scientific (UK) ESCALAB Xi^+ system.

2.6. Evaluation and CPX detection using CD-AgNPs/GCE

Electrochemical investigations were carried out using a PGSTAT 302 N Potentiostat/Galvanostat system (Metrohm, Switzerland). The system is configured with a standard three-electrode cell. The setup comprised



Fig. 1. Illustration for CDs and CD-AgNPs synthesis from Thai basil leaves.

an Ag/AgCl (3 M KCl) as the reference electrode, a platinum wire (diameter: 1 mm) acting as the counter electrode, and the working electrode consisted of either bare GCE or CD-AgNPs/GCE. The electrochemical impedance spectra (EIS) were employed to analyze the R_p at the interfaces of both the bare and modified GCE in a 0.1 M KCl electrolyte using 1 mM $\text{Ru}(\text{NH}_3)_6\text{Cl}_2$ as the standard probe within the frequency range of 0.01 to 100,000 Hz.

CV was recorded to investigate A_e over a scan rate range of 5 to 500 mV s^{-1} and to compare the anodic/cathodic peak current (I_{pa}/I_{pc}) using bare and modified GCE in a 0.1 M KCl electrolyte solution with 1 mM $\text{Ru}(\text{NH}_3)_6\text{Cl}_2$ as the redox probe (fixed scan rate of 50 mV s^{-1}) maintained at 25 °C. Furthermore, CV analysis was employed to (i) study the influence of electrolyte pH adjusted from 3 to 9 in 0.1 M phosphate buffer solution (PBS) during CPX sensing at 25 °C, (ii) record the I_{pa} response of CPX using bare GCE and CD-AgNPs/GCE electrodes under optimized pH conditions at 25 °C, (iii) evaluate electrokinetic parameters such as the total electron count associated with the CPX redox mechanism (n), k^0 , and E^0 in (0.1 M) PBS buffer (25 °C, pH 5), and (iv) analyze the Γ^* of CPX under the same electrolyte and temperature conditions.

SWV measurements were conducted to (i) evaluate the influence of deposition variables, namely DP (deposition potential), and DT (deposition time), on CPX detection in (0.1 M) PBS buffer (pH 5), (ii) establish a calibration curve for CPX quantification across different concentrations (0–200 μM) and determine the LOD, LOQ, and sensitivity in (0.1 M) PBS buffer (25 °C, pH 5), (iii) examine the reproducibility, stability, and selectivity (potential interfering substances: phenylbutazone, chloroquine, amoxicillin, chloramphenicol, and sulfamethoxazole) of the fabricated electrode during CPX sensing, and (iv) analyze and quantify CPX in environmental samples such as tap water, and river water under identical experimental conditions. Ahead of the measurements, water samples were passed through 0.2 μm membrane filters to eliminate particulate matter. All SWV tests were performed at a fixed frequency of 25.0 Hz, an amplitude of 0.02 V, a step potential of 0.002 V, and a scan rate of 50 mV s^{-1} .

3. Results and discussion

3.1. Characterization of CDs and CD-AgNPs

The optical properties of the synthesized CDs and CD-AgNPs were evaluated using UV–Vis and fluorescence spectroscopy (Fig. 2a and b). The UV–Vis spectrum of the CDs (Fig. 2a) exhibited a prominent absorption peak at 276 nm, which can be assigned to the π – π^* transition of C=C bonds. Upon the addition of Ag^+ ions, a new absorption band appeared at 431 nm, characteristic of the surface plasmon resonance of AgNPs. This change was also accompanied by a visible color shift of the

solution from light brown to dark red, indicating the successful formation of the CD-AgNPs (Fig. 2a, inset). Fig. 2b illustrates the fluorescence spectra of the CDs recorded at excitation wavelengths ranging from 270 to 390 nm. A redshift in the emission peak was observed with increasing excitation wavelength. The emission spectrum of the CDs shows a maximum of 416 nm when excited at 330 nm. The inset photograph demonstrates that the CDs' aqueous solution emits bright blue fluorescence when exposed to a 365 nm UV lamp and shows a yellowish hue under normal lighting conditions [30,31]. The photoluminescent characteristics of the synthesized CDs align with recent advances in engineered luminogenic nanomaterials, where tunable optical properties have been leveraged for multifunctional applications [32], suggesting that the present basil-derived CDs could be further developed toward dual-mode optical–electrochemical sensing platforms.

The morphology and particle size of the prepared CDs were determined using FETEM. FETEM image and particle size distribution graph (Fig. 3a and inset image) reveal a well-dispersed formation of CDs with a spherical morphology and an average size of about 5 nm. A total of $n = 20$ particles were measured using ImageJ 1.52a software, and the distribution curve was fitted with a Gaussian function to obtain the mean particle diameter. The high-resolution TEM (HRTEM) image (Fig. 3b) reveals a lattice fringe spacing of 0.25 nm, which corresponds to the (100) lattice plane of graphene [33]. Fig. 3c depicts the selected area electron diffraction (SAED) pattern, displaying a broad, diffused ring, which is indicative of the predominantly amorphous structure of the CDs [34]. The FETEM images of the CD-AgNPs showed that the NPs were roughly spherical, evenly distributed, and well-dispersed (Fig. 3d) and had a mean diameter of 20.24 ± 2.5 nm as per the size distribution curve (Fig. 3d, inset image). Similarly, to determine the mean particle size, a total of 60 particles ($n = 60$) were analyzed using ImageJ 1.52a software, and the resulting size distribution was fitted with a Gaussian function. HRTEM analysis further revealed an interplanar spacing of 0.235 nm, which matches the (111) plane of AgNPs (Fig. 3e). The SAED pattern displayed several bright concentric rings corresponding to the (111), (200), (220), (311), and (222) planes, confirming the face-centered cubic crystalline structure of the AgNPs (Fig. 3f). These crystallographic features were in good agreement with the XRD results, validating the structural integrity of the synthesized NPs.

XRD analysis was conducted to further elucidate the crystalline nature of CD-AgNPs. As illustrated in Fig. 4a, the CDs exhibit a broad diffraction peak centered at approximately $2\theta \approx 21.5^\circ$ (curve b), which is indicative of the (002) plane typical of amorphous carbon. This observation is consistent with the structural details observed in the SAED pattern [34]. Curve a, representing CD-AgNPs, exhibited distinct XRD peaks at 2θ positions of 38° , 44° , 64° , and 78° , indexed to the (111), (200), (220), and (311) lattice planes of AgNPs (JCPDS file No.: 00–004–

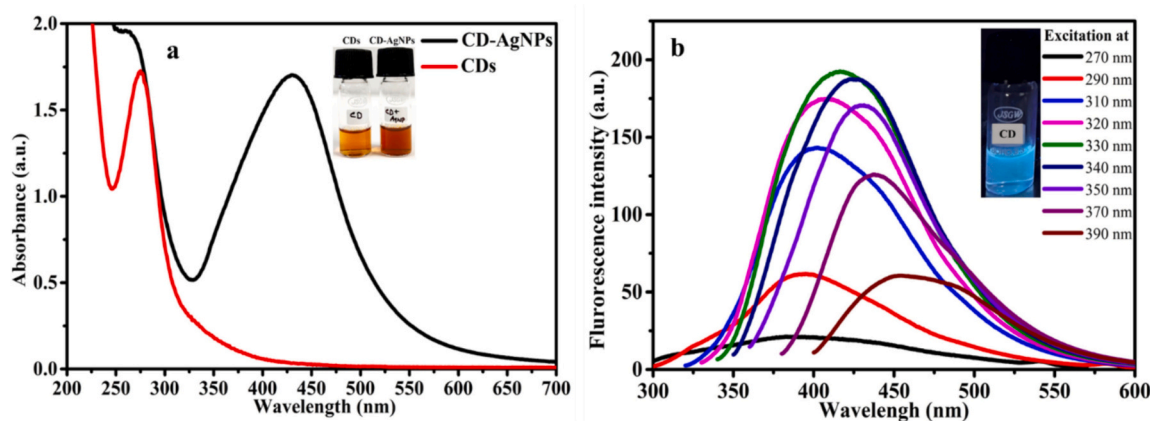


Fig. 2. (a) Absorption spectra (UV–Vis) of synthesized CDs and CD-AgNPs (inset: optical photograph of CDs and CD-AgNPs under visible light) and (b) fluorescence emission spectra recorded at different excitation wavelengths (270–390 nm) (inset: optical photograph of CDs under 365 nm UV light).

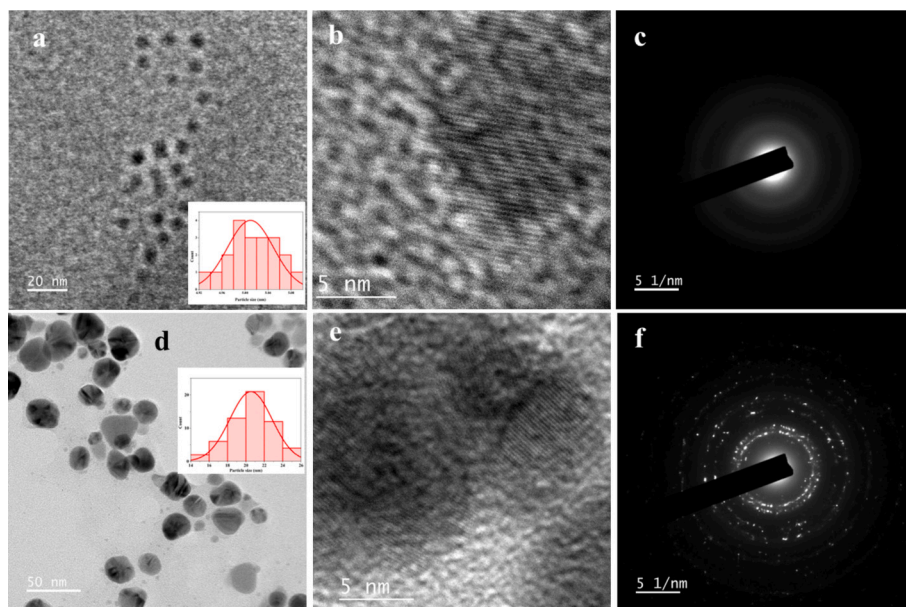


Fig. 3. (a) FETEM image, (b) HRTEM micrograph, (c) SAED pattern of synthesized CDS, (d) FETEM image, (e) HRTEM micrograph, and (f) SAED pattern of synthesized CD-AgNPs.

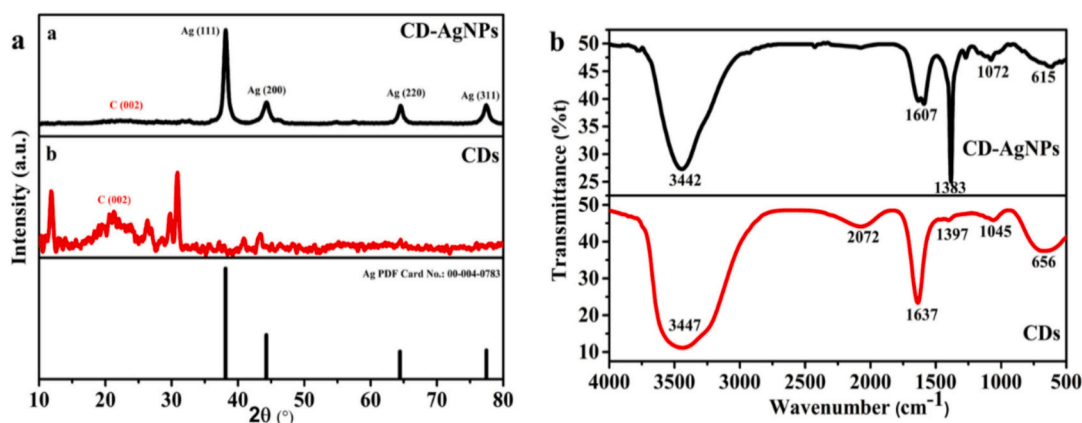


Fig. 4. (a) XRD profiles of (a) CD-AgNPs along with the (b) CDs, (b) FTIR spectra corresponding to CD-AgNPs and CDs.

0783), confirming their crystalline nature. Additionally, the peak at 21.5° , which is characteristic of the disordered carbon in CDs, showed a noticeable decrease in intensity after forming the CD-AgNPs. This attenuation is attributed to the dominant crystalline reflections of AgNPs and the relative dilution of the CD phase within the hybrid structure, rather than structural alteration of the carbon framework [35]. The crystallite size of CD-AgNPs was calculated using Scherrer's Eq. [36], expressed as follows:

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (1)$$

In this equation, D represents the average crystallite size, λ denotes the wavelength of the X-ray radiation (0.154 nm), θ corresponds to the Bragg diffraction angle ($2\theta/2$), and β is the full width at half maximum expressed in radians. Using this equation, the average crystallite size was calculated to be 20.53 nm (Table S1, Supplementary Material), which is in good agreement with the particle size calculated from FETEM analysis. This nanoscale crystallinity of the CD-AgNPs provides an enlarged surface area, enhancing the availability of active sites and accelerating electron transfer kinetics, which results in enhanced peak current response and improved sensitivity.

The surface chemistry of both CDs and CD-AgNPs was analyzed using FTIR to identify the existing functional groups (Fig. 4b). The broad absorption peak at around 3447 cm^{-1} can be ascribed to the stretching modes of O—H and N—H bonds, suggesting the presence of hydroxyl and amine groups on the surface. In the spectrum of the CDs (curve b), a clear band at 1637 cm^{-1} can be ascribed to amide (—CONH—) linkages, while the peak observed near 2072 cm^{-1} is related to the C—H stretching of aldehyde groups. The signals at 1397 cm^{-1} and 1045 cm^{-1} are attributed to O—H bending and C—O stretching vibrations, respectively. The band appearing around 656 cm^{-1} may correspond to flavonoid-related structures, most likely derived from the plant-based carbon precursor. After the incorporation of AgNPs, a marked reduction in the intensity of peaks at 3447 cm^{-1} (secondary amine) and 1637 cm^{-1} (amide) was evident, confirming the successful formation of the CD-AgNPs composite. Additionally, the band at 2072 cm^{-1} nearly disappeared, while a new prominent peak at 1383 cm^{-1} emerged, which can be associated with C—H bending vibrations of oxidized aldehyde species [10,35]. These results correspond to those obtained from the XPS spectra of CD-AgNPs.

To determine the surface chemistry and elemental composition of CD-AgNPs, XPS analysis was conducted. Fig. 5a presents the complete

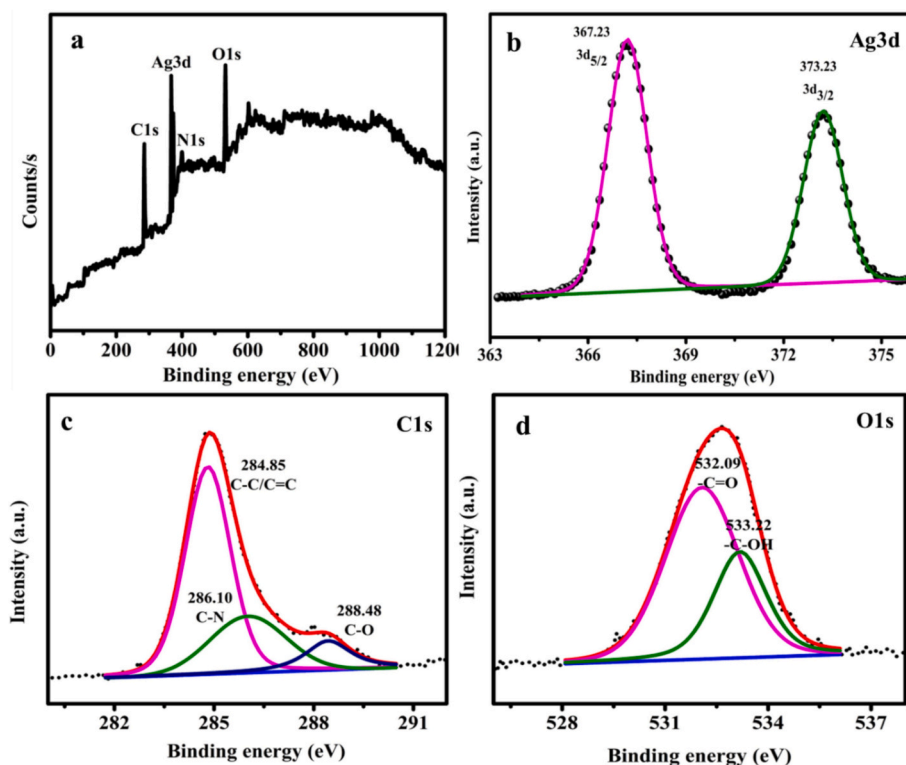


Fig. 5. (a) XPS survey spectrum and high-resolution XPS spectra of (b) Ag3d, (c) C1s, and (d) O1s for the synthesized CD-AgNPs.

XPS survey of CD-AgNPs, revealing the presence of C1s at 285.2 eV, Ag3d at 367.2 eV, N1s at 400.2 eV, and O1s at 532.2 eV. Two distinct peaks appear in the Ag3d spectrum (Fig. 5b) at 367.23 and 373.23 eV, attributed to the 3d_{5/2} and 3d_{3/2} states of silver, confirming the presence of metallic silver. The C1s spectrum (Fig. 5c) displays binding energy peaks at 288.48, 286.1, and 284.85 eV, which are assigned to C—O, C—N, and C—C/C=C groups, respectively. In the O1s spectrum (Fig. 5d), peaks at 532.09 eV and 533.22 eV correspond to —C=O and —C—OH groups, respectively [35,37].

3.2. Electrochemical characterization of CD-AgNPs/GCE

3.2.1. EIS analysis

EIS analysis was employed to evaluate the electron transport resistance of both bare and CD-AgNPs/GCE electrodes. The measurements were conducted in 0.1 M KCl supplemented with 1 mM Ru(NH₃)₆Cl₂ as the redox probe, over a frequency range of 10^{−2}–10⁵ Hz at 25 °C. KCl solution served as the supporting electrolyte to ensure consistent ionic strength throughout the experiment. [Ru(NH₃)₆]Cl₂ was employed as the redox probe owing to its excellent electrochemical properties, such as a stable one-electron transfer mechanism and clearly defined redox peaks. Fig. 6a presents the EIS spectra for both the bare GCE and CD-AgNPs/GCE together with the Randles equivalent circuit (inset), highlighting key electrokinetic parameters including R_p, constant phase elements (CPEs), and solution resistance (R_s), which represents the non-ideal capacitive behavior of the electrode interface. In a Nyquist plot, the diameter of the semicircle reflects the R_p, where a larger semicircle corresponds to a higher R_p value [38]. For the unmodified GCE, R_p was found to be approximately 2842 kΩ. After modification with CD-AgNPs, this value decreased markedly to about 851 kΩ, signifying a notable improvement in electron transfer efficiency and overall conductivity of the modified electrode surface. The R_s was 0.128 kΩ for the bare GCE and 0.141 kΩ after CD-AgNPs coating. This minor change in R_s can be attributed to slight variations in the spatial positioning of the electrodes during repeated experiments or alterations in electrode surface

morphology after CD-AgNPs deposition, which can affect ion mobility within the electrolyte. Nevertheless, the difference in R_s is minimal when compared to the change observed in R_p values. Variations in the CPE are primarily attributed to differences in the conductivity and surface characteristics of the electrodes. A detailed comparison of these electrochemical parameters for both electrodes is provided in Table 1.

3.2.2. Surface area for bare and CD-AgNPs/GCE

The A_e of both the bare GCE and CD-AgNPs/GCE were assessed through CV conducted at varying scan rates ranging from 5 to 500 mV s^{−1}. Measurements were performed in a 0.1 M KCl electrolyte solution containing 0.001 M [Ru(NH₃)₆]Cl₂ as the redox probe, at a constant temperature of 25 °C (as shown in Fig. 6b and c). The A_e was calculated using the Randles–Ševčík equation (Eq. 2) [39]:

$$I_{pa} = (2.69 \times 10^5) n^{3/2} A_e D^{1/2} \nu^{1/2} C_0 \quad (2)$$

Where I_{pa} is the anodic peak current (A), C₀ is the concentration of the redox species (mol cm^{−3}), ν is the scan rate (V s^{−1}), n is the number of electrons involved in the redox process (n = 1), D is the diffusion coefficient (8.43 × 10^{−6} cm² s^{−1}) and A_e is the electroactive surface area (cm²). The linear relationship between I_{pa} and the $\nu^{1/2}$ is depicted in Fig. 6d, where the slopes for the bare GCE and CD-AgNPs/GCE were found to be 31.228 μA·V^{−1/2}·s^{1/2} (R² = 0.998) and 68.219 μA·V^{−1/2}·s^{1/2} (R² = 0.999), respectively. By equating these slope values with the Randles–Ševčík equation, the calculated A_e were 0.040 cm² for the bare GCE and 0.087 cm² for the CD-AgNPs/GCE. This represents an approximate 54.02% increase in electroactive A_e upon CD-AgNPs deposition, providing more active sites for CPX interaction and significantly improving the sensor's detection capabilities.

3.3. Optimization of pH, and deposition parameters

3.3.1. Effect of pH

Fig. 7a presents the CV profiles for the electrocatalytic detection of

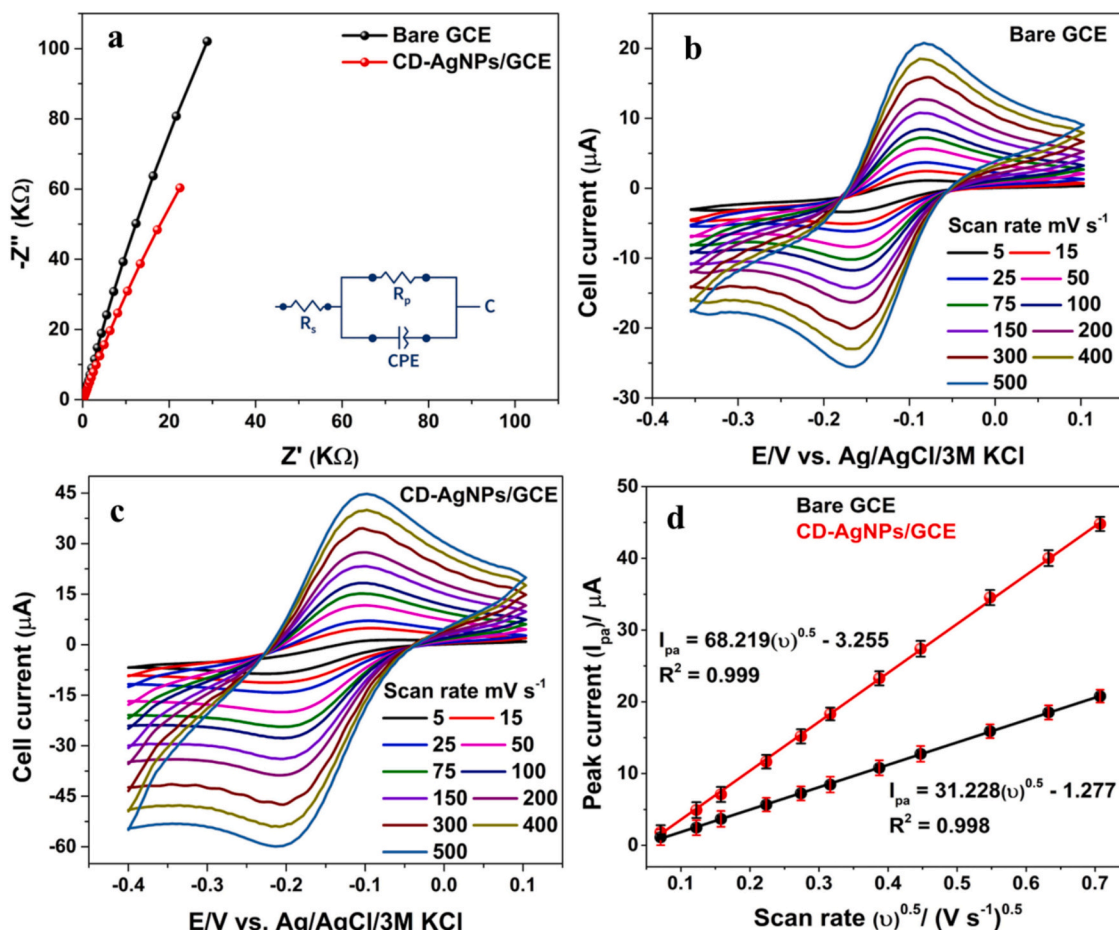


Fig. 6. (a) EIS results illustrating the electron transport resistance of bare and CD-AgNPs/GCE electrodes, along with the corresponding Randles equivalent circuit (inset); (b) and (c) CV of bare and CD-AgNPs/GCE recorded at varying scan rates (5–500 mV s^{-1}), and (d) relationship between the I_{pa} and the $v^{0.5}$ for both electrodes.

Table 1

Fitting Parameters from Nyquist Plots for bare GCE and CD-AgNPs/GCE.

Working electrode	R_s (k Ω)	R_p (k Ω)	CPE (μMho)
Bare GCE	0.128 ± 0.005	2842 ± 4.59	14.3 ± 0.04
CD-AgNPs/GCE	0.141 ± 0.003	851 ± 2.65	21.1 ± 0.03

CPX (60 μM) using CD-AgNPs/GCE in PBS (0.1 M) in the pH domain of 3 to 9, executed under a scan rate of 50 mV s^{-1} and a temperature of 25 $^\circ\text{C}$. A clear shift in the oxidation peak potential (E_p) toward less positive values with increasing pH suggests the involvement of protons in the redox process of CPX, as shown in Fig. 7a [11,38,40]. Among all the examined pH conditions, the oxidation peak current reached its maximum at pH 5, suggesting that this environment offered the most favorable conditions for achieving high detection sensitivity and electrocatalytic performance (Fig. 7b) [41]. Therefore, a PBS buffer (0.1 M) of pH 5 was chosen for subsequent electrochemical studies. The Nernst equation was then employed to evaluate the ratio of electron-to-proton (n/m) participating in the CPX detection process [38].

$$\frac{dE_p}{dpH} = -2.303 \frac{mRT}{nF} \quad (3)$$

In this context, T stands for absolute temperature (K), R indicates the universal gas constant (8.314 $\text{J mol}^{-1} \text{K}^{-1}$), and F denotes Faraday's constant (96,485 C mol^{-1}). The plot of E_p versus pH (Fig. 7c) showed a clear linear trend, giving a slope of approximately $-0.0571 \text{ V per pH unit}$ ($R^2 = 0.991$). Using this slope value in the Nernst equation, the ratio

of electrons to protons (n/m) involved in the redox process was determined to be around 1.04 (≈ 1), suggesting that the electrochemical sensing of CPX proceeds with nearly equal participation of electrons and protons. Additionally, Fig. 7d compares the CV responses of CPX detection using bare GCE, and the fabricated CD-AgNPs/GCE under identical conditions (PBS (0.1 M) at pH 5, 60 μM CPX, 50 mV s^{-1} , and 25 $^\circ\text{C}$), demonstrating the superior electrocatalytic performance of the CD-AgNPs/GCE. The CD-AgNPs/GCE demonstrated a notable enhancement in electrocatalytic performance for CPX detection, with a 42.1% notable improvement in I_{pa} as compared to the bare GCE.

3.3.2. Effect of deposition parameters

SWV was executed in PBS buffer (0.1 M) (pH 5) at 25 $^\circ\text{C}$ containing 60 μM CPX and a scan rate of 50 mV s^{-1} to evaluate how parameters governing deposition—specifically potential (DP vs. Ag/AgCl) and DT—influence the I_{pa} for CPX detection using CD-AgNPs/GCE (Fig. S1 provided in Supplementary Material). To analyze the impact of DP on determining the electrode's sensitivity for CPX detection, the potential was varied between -0.1 V and -0.7 V while maintaining DT at 90 s (Fig. S1a). A steady enhancement of the I_{pa} was noted as the DP shifted from -0.1 V to -0.5 V , which can be attributed to the improved adsorption of CPX molecules on the CD-AgNPs/GCE surface (Fig. S1b). When the potential exceeded -0.5 V , the current response began to drop, likely due to hydrogen gas formation at more negative potentials, which can hinder the electrochemical reaction and reduce signal intensity [42]. Therefore, $-0.5 \text{ V vs. Ag/AgCl}$ was chosen as the optimal DP for achieving maximum detection sensitivity toward CPX.

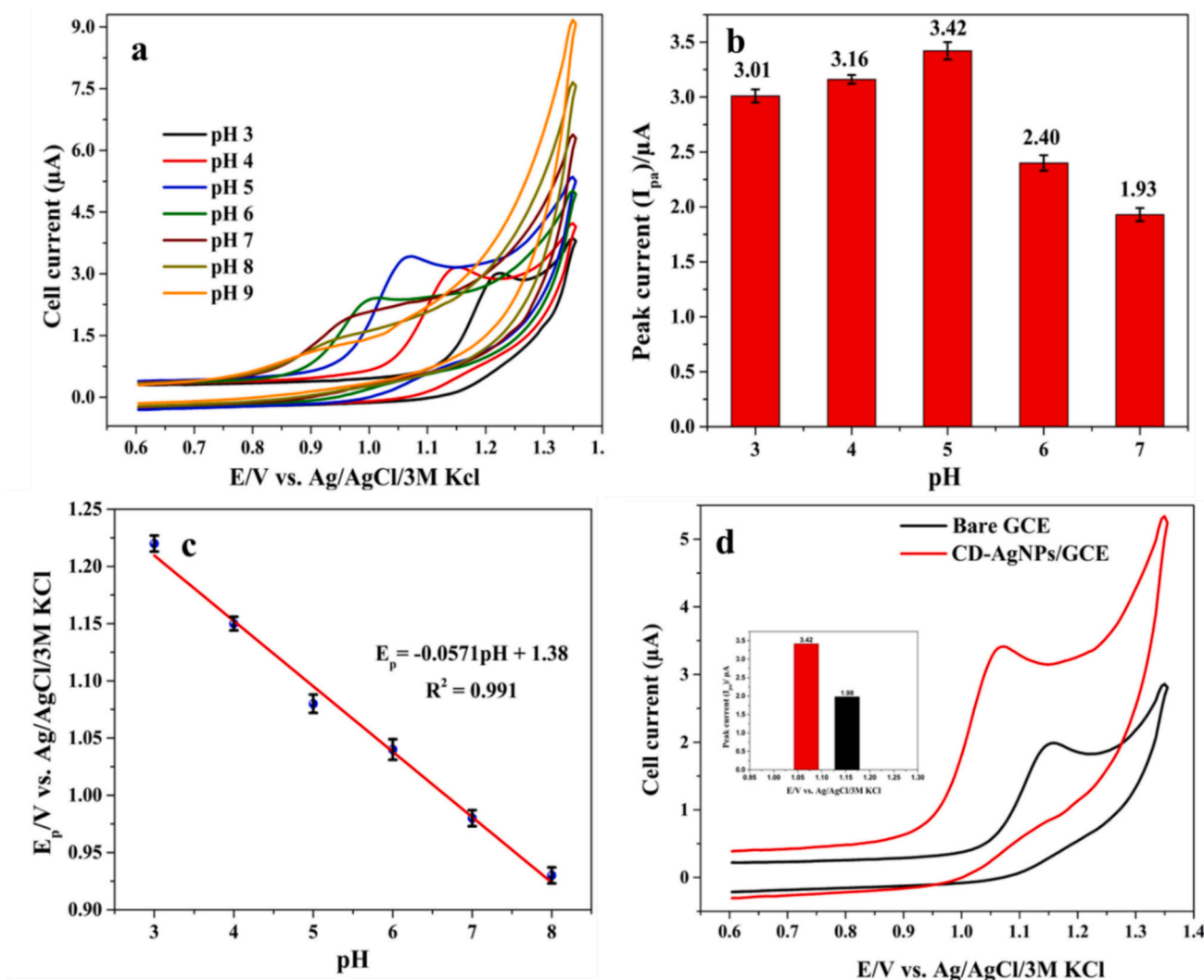


Fig. 7. (a) CV profiles of CPX recorded at varying pH levels (3–9) in PBS (0.1 M) using the CD-AgNPs/GCE electrode, (b) variation of I_{pa} with pH, (c) relationship between E_p and pH, and (d) CV responses for 60 μM CPX detection using bare GCE, and CD-AgNPs/GCE in PBS (0.1 M) (pH 5, 25 °C) at a scan rate of 50 mV s⁻¹, with the inset showing a comparison of anodic peak currents obtained from both the electrodes.

Figs. S1c and S1d depict the effect of varying DT on the oxidation peak of CPX at a fixed potential of -0.5 V. A gradual rise in the I_{pa} was observed as the DT increased from 30 s to 90 s, suggesting enhanced adsorption or accumulation of CPX on the CD-AgNPs/GCE surface. Beyond 90 s, the increase in peak current became negligible, suggesting that the electrode surface might have reached saturation or that an equilibrium had formed between the adsorbed CPX and the electrolyte. This plateau could also result from the limited availability of CPX ions in the fixed-concentration PBS medium, reducing the number of molecules participating in the redox process. Therefore, a DT of 90 s was selected as the optimal condition for subsequent electrochemical studies involving CPX detection.

3.4. Electrochemical CPX sensing at CD-AgNPs/ GCE

3.4.1. Assessment of electro-kinetic behaviors

To evaluate the electro-kinetic behavior associated with the oxidation and surface adsorption of CPX on the CD-AgNPs/GCE, CV was conducted over a range of scan rates from 5 to 500 mV s⁻¹ in 0.1 M PBS (pH 5) at 25 °C using a CPX concentration of 60 μM (as shown in Fig. 8a). The slope of the linear plot between the I_{pa} and the $\nu^{0.5}$, calculated as 23.951 μA·V^{-1/2}·s^{1/2} with a R^2 of 0.984, suggests that the oxidation process of CPX at the CD-AgNPs/GCE is predominantly governed by

diffusion control reaction (Fig. 8b). Accordingly, the Laviron equation (eq. 4) and an additional equation (eq. 5) were applied to evaluate the electrokinetic parameters as well as the adsorption behavior of CPX for the diffusion-controlled process.

$$E_p = E^0 - \left(\frac{2.303RT}{\alpha nF} \right) \cdot \log \left(\frac{RTk^0}{\alpha nF} \right) + \left(\frac{2.303RT}{\alpha nF} \right) \log \nu \quad (4)$$

$$I_{pa} = (n^2 F^2 A_e \nu \Gamma^*) / (4RT) \quad (5)$$

In these expressions, E_p and E^0 refer to the observed and standard oxidation potentials (V), respectively, k^0 is the electron transfer rate constant (s⁻¹), ν is the scan rate (Vs⁻¹), n represents the number of electrons involved in the redox reaction, Γ^* is the surface coverage or adsorbed amount of CPX on the electrode (mol·cm⁻²), R is the universal gas constant (8.314 J·mol⁻¹·K⁻¹), F is Faraday's constant (96,485 C·mol⁻¹), α is the charge transfer coefficient (assumed to be 0.5), T is the absolute temperature in Kelvin, A_e is the electroactive surface area of the electrode (cm²). The E^0 was obtained by extrapolating the E_p versus ν plot at $\nu = 0$, giving a value of 1.077 V. As shown in Fig. 8c, E_p exhibits a linear correlation with $\log \nu$, yielding a slope of 0.050 V and an intercept of 1.207 V ($R^2 = 0.985$). Using $\alpha = 0.5$ and inserting the values of E^0 (1.077 V), slope (0.050 V), and intercept (1.207 V) into eq. 4, the calculated n and k_0 were 2.37 and 0.1157 s⁻¹,

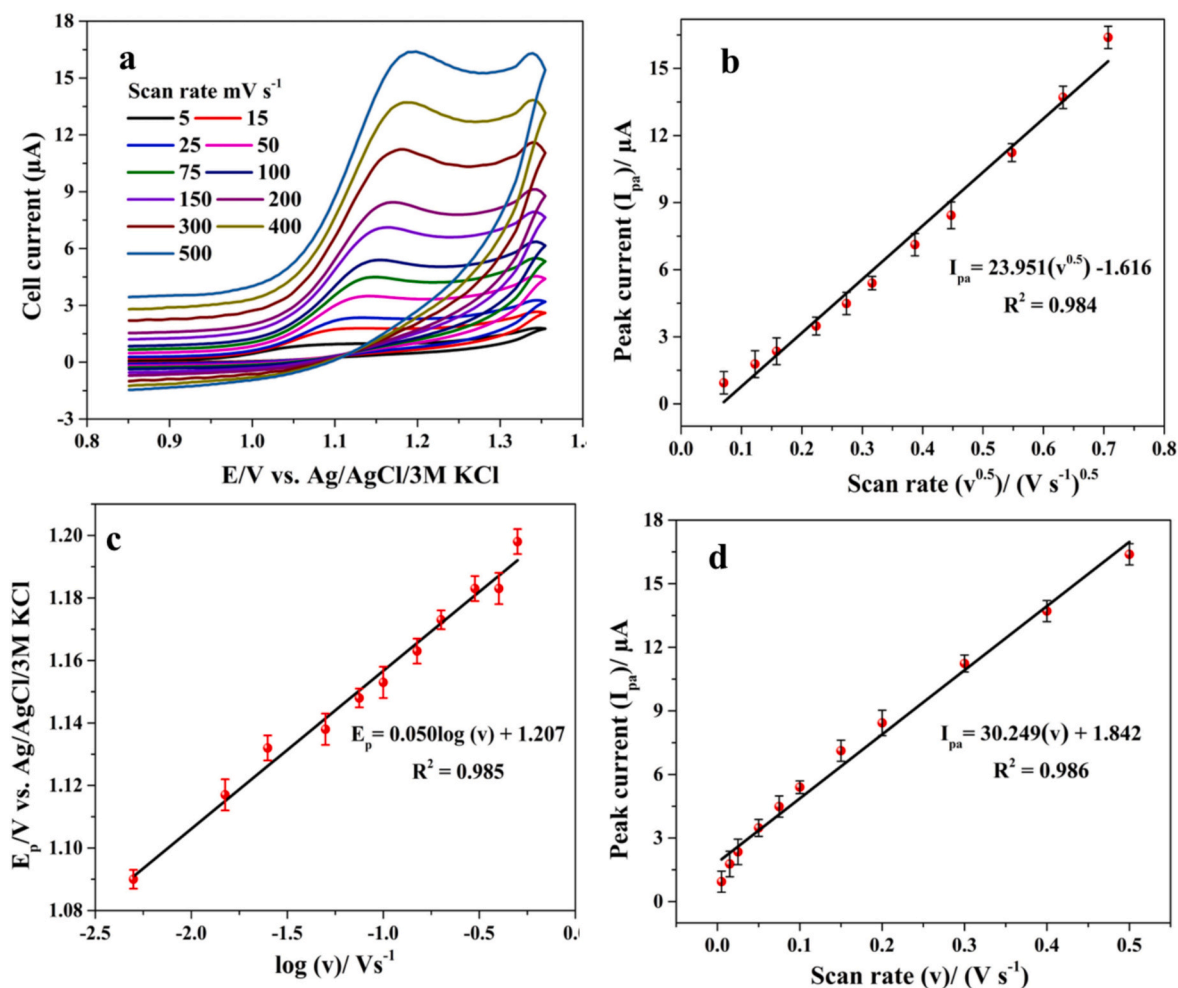


Fig. 8. (a) CV measurements were performed at varying scan rates (5–500 mV s⁻¹) to evaluate the electrokinetic properties of CPX, (b) plot of I_{pa} versus $v^{0.5}$, (c) variation of E_p with $\log v$, and (d) relationship between I_{pa} and v .

respectively. This indicates that the electrochemical sensing of CPX involves the transfer of two electrons [43]. Fig. 8d shows a linear relationship between I_{pa} and v , with a slope of 30.249 $\mu\text{A}\cdot\text{V}^{-1}\cdot\text{s}$ ($R^2 = 0.986$). By substituting n (2.37), A_e (0.087 cm^2 , determined from eq. 2), and the slope value (30.249 $\mu\text{A}\cdot\text{V}^{-1}\cdot\text{s}$) into eq. 5, Γ^* was calculated as $6.593 \times 10^{-11} \text{ mol}\cdot\text{cm}^{-2}$.

In the present analysis, α was assumed to be 0.5, which is commonly adopted for quasi-reversible systems when the electron-transfer process is symmetric, and α cannot be independently extracted with high accuracy from the available data. In diffusion-controlled systems, simultaneous determination of α , n , and k^0 from Laviron's equation may introduce significant uncertainty due to parameter interdependence. Following established electrochemical practice and previously reported methodologies, the values of n and k^0 were determined from the slope and intercept of the E_p vs $\log v$ plot using Laviron's equation, considering $\alpha = 0.5$. The E^0 was estimated by extrapolating E_p versus v to $v \rightarrow 0$, as described in the literature [44].

3.4.2. CPX sensing over CD-AgNPs/GCE

Fig. 9a presents the SWV response obtained from CD-AgNPs/GCE in PBS buffer (0.1 M, pH 5) at 25 °C, using optimized deposition conditions of -0.5 V (DP) and 90 s (DT). The experiment was carried out across a range of CPX concentrations from 0 to 200 μM . The resulting plot of I_{pa} versus CPX concentration exhibited two distinct linear trends: one within the 0.25–40 μM range and another between 40 and 200 μM (Fig. 9b). This shift in linearity is attributed to a transition in the

electrochemical mechanism—from diffusion-controlled oxidation at lower concentrations to surface-controlled processes at higher concentrations, likely due to saturation or dense accumulation of CPX molecules on the electrode surface [45]. The LOD and LOQ for CPX sensing with the modified CD-AgNPs/GCE were calculated using Eqs. (6) and (7), respectively [45].

$$\text{LOD} = 3S_d/m \quad (6)$$

$$\text{LOQ} = 10S_d/m \quad (7)$$

In this study, m denotes the slope of the calibration curve obtained from the plot of I_{pa} versus CPX concentration, whereas S_d represents the standard deviation of the blank response, determined to be 0.0026. This approach is widely adopted in electrochemical sensing studies and has been implemented in several previously reported works [38,46]. However, alternative methods for LOD estimation have also been reported in the literature [47]. As illustrated in Fig. 9b, the I_{pa} vs. CPX concentration plot exhibited two distinct linear slopes: 0.0694 $\mu\text{A}\cdot\mu\text{M}^{-1}$ ($R^2 = 0.988$) for the 0.25–40 μM range and 0.0241 $\mu\text{A}\cdot\mu\text{M}^{-1}$ ($R^2 = 0.989$) for the 40–200 μM interval. The LOD and LOQ values derived from the calibration slopes were 0.1124 and 0.3746 μM for the lower range, and 0.3237 and 1.079 μM for the higher range, respectively. Sensitivity, calculated by dividing the slope by the electrode's active area ($A_e = 0.087 \text{ cm}^2$) [46], was found to be 0.7977 $\mu\text{A}\cdot\mu\text{M}^{-1}\cdot\text{cm}^{-2}$ (0.25–40 μM) and 0.2770 $\mu\text{A}\cdot\mu\text{M}^{-1}\cdot\text{cm}^{-2}$ (40–200 μM). These results demonstrate the sensor's high analytical performance over a broad concentration range.

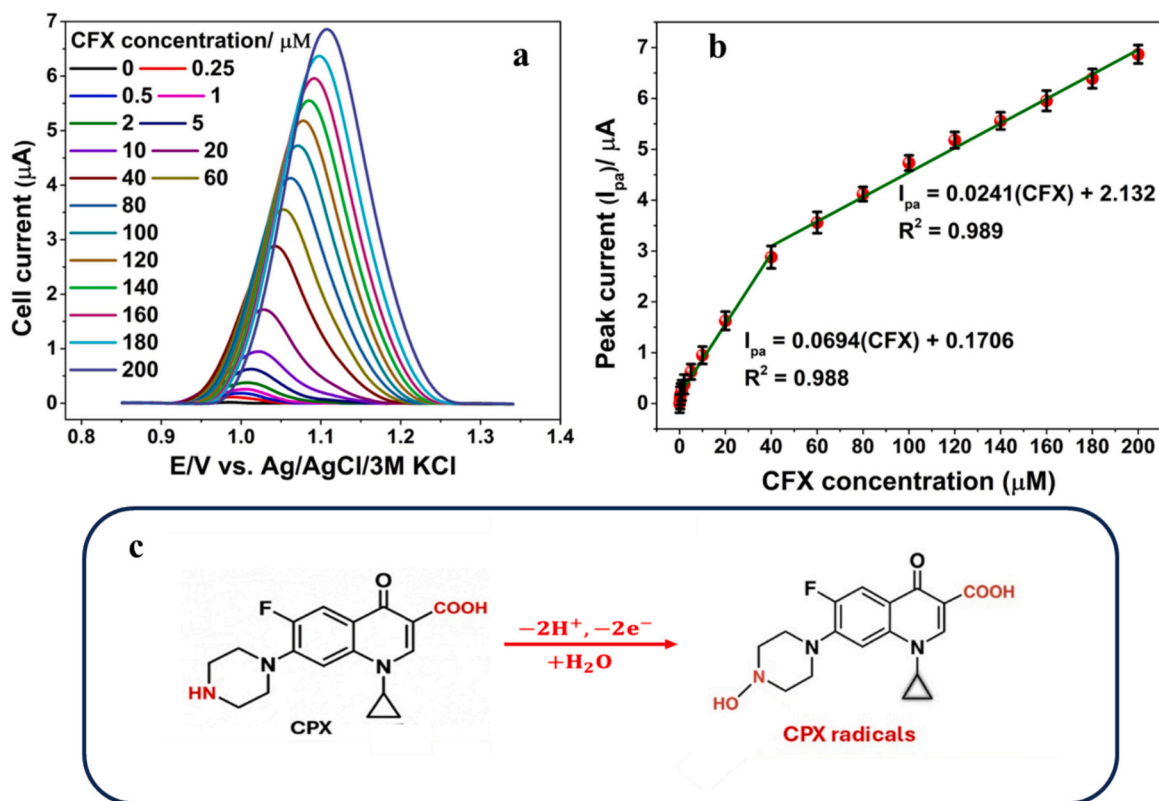


Fig. 9. (a) SWV responses of CD-AgNPs/GCE for CPX concentrations ranging from 0 to 200 μM, (b) corresponding I_{pa} versus CPX concentration calibration plot, and (c) schematic depicting the redox reaction pathway responsible for CPX detection.

3.4.3. Electrocatalytic detection process and CPX interaction with CD-AgNPs/ GCE

The integration of CD-AgNPs on the GCE surface improved its electrochemical performance by enlarging the surface area, facilitating charge transfer, and enhancing analyte diffusion [48]. The catalytic nature of AgNPs further accelerated redox reactions, leading to efficient CPX detection. Fig. 9c illustrates the oxidation of CPX to its radical form, consistent with a two-electron, two-proton transfer mechanism. This electron transfer aligns with the number of electrons previously calculated using the Laviron equation (Eq. 4), validating the electrochemical findings. This interpretation aligns with the two-step interfacial electron-transfer model reported by Lin et al. [43], where sequential electron-transfer processes govern electrode kinetics and peak behavior. The formation of the CPX radical at the CD-AgNPs/GCE surface confirms

that oxidation of CPX takes place during the sensing process. Hence, the proposed detection approach is based on monitoring the electrochemically generated CPX radical.

3.4.4. Stability and reproducibility of CD-AgNPs/GCE during CPX detection

To evaluate the long-term stability and reproducibility of the CD-AgNPs/GCE during CPX detection, SWV was performed in PBS buffer (0.1 M, pH 5) at 25 °C. The experiments were conducted using optimized deposition conditions—DP of −0.5 V, DT of 90 s, and a scan rate of 50 mV s^{−1}—at a fixed CPX concentration of 60 μM. For the stability assessment, a single CD-AgNPs/GCE electrode was employed repeatedly in the same electrolyte solution across 10 consecutive measurement cycles. In the reproducibility study, three freshly prepared CD-AgNPs/

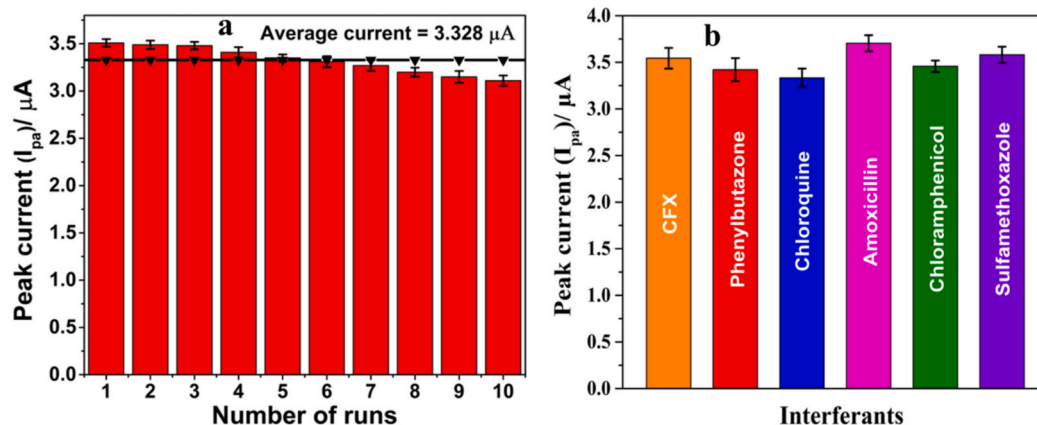


Fig. 10. (a) Stability evaluation of the CD-AgNPs/GCE and (b) its selectivity analysis in the presence of potential intrusive species for instance phenylbutazone, chloroquine, amoxicillin, chloramphenicol, and sulfamethoxazole.

GCE electrodes were tested individually in four separate PBS solutions (each containing 60 μM CPX) under identical experimental conditions. Fig. 10a illustrates a gradual decline in the I_{pa} of CPX with successive measurements. Specifically, the current dropped by only 11.4% after 10 cycles, indicating strong retention of electrocatalytic performance by the CD-AgNPs/GCE, particularly when compared to earlier reports [11]. The observed reduction in peak current may result from the progressive accumulation of CPX molecules on the electrode surface, which can hinder further electron transfer [46]. In terms of reproducibility, Fig. S2 highlights the excellent consistency of the sensor's response, showing a relative standard deviation (RSD) of just 4.87%, which falls well within the acceptable range for analytical applications. Table S2 summarizes the performance metrics of the CD-AgNPs/GCE for CPX sensing and benchmarks them against recently reported sensors, revealing that the developed platform delivers superior detection capabilities over many previously published methods.

3.5. Selectivity of CPX detection over CD-AgNPs/GCE

Fig. 10b illustrates the impact of potential interfering substances such as phenylbutazone, chloroquine, amoxicillin, chloramphenicol, and sulfamethoxazole on the electrochemical detection of CPX using CD-AgNPs/GCE under optimal conditions. The experiment involved additions of CPX (60 μM) and the interferants (40 μM each) altogether over the CD-AgNPs/GCE. Importantly, the presence of these interferants did not significantly influence the sensor's output when the tolerance limit was considered as an $\pm 5\%$ error. This lack of interference confirms that the CD-AgNPs/GCE offers high selectivity toward CPX detection, even in the presence of structurally similar compounds. Beyond interference testing, rational recognition-site engineering has been demonstrated as an effective strategy to enhance molecular specificity in electrochemical sensors, as reported by Li et al. [49], suggesting a potential direction for further improving selectivity of the CD-AgNP-based platform in complex matrices.

3.6. Quantification of CPX in aqueous samples

To assess the real-world applicability of the CD-AgNPs/GCE sensor for detecting CPX, SWV was carried out using a PBS buffer (0.1 M, pH 5) solution mixed with water samples from tap and river, each spiked with varying concentrations of CPX (5–80 μM). The experiments were conducted under optimized conditions (DP: -0.5 V, DT: 90 s), and at a temperature of 25 $^{\circ}\text{C}$. The aquatic specimens were sourced from the Chao Phraya River and the laboratory provisions at Chulalongkorn University, Bangkok, Thailand. Pre-analysis, all aquatic specimens were refined using 0.2 μm membranes to remove particulate matter and minimize the risk of electrode surface contamination, ensuring reliable performance of the CD-AgNPs/GCE during electrochemical quantification. The analytical execution of the CD-AgNPs/GCE sensor was evaluated by detecting known concentrations of CPX spiked into tap and river

water samples, utilizing the calibration curve derived from Fig. 9b. These results were then cross-validated using HPLC. A summary of the sensor's performance, along with its correlation to HPLC outcomes, is provided in Table 2. The recovery rates for CPX in both tap and river water were approximately 100%, indicating high accuracy. Deviations in CPX concentration measurements between the CD-AgNPs/GCE sensor and HPLC were maximum—6.4% for tap water (with a maximum RSD of 5.2%) and 8.1% for river water (maximum RSD of 6.8%). The variation between the spiked concentrations and the experimentally determined values in the real samples was statistically evaluated using a two-tailed Student's *t*-test [50]. The test ($n = 3$, $n - 1 = 2$ degrees of freedom, 95% confidence level) indicated that the calculated *t* values were lower than the critical *t* value (4.303), confirming that no statistically significant difference exists between the added and measured concentrations (Table 2). These findings demonstrate that the developed CD-AgNPs/GCE sensor exhibits excellent reliability, making it well-suited for on-site monitoring and assessment of CPX in complex aquatic environmental samples. Recent advances in self-powered sensing platforms underscore the potential of the proposed CD-AgNP-modified electrode for autonomous, in-field monitoring of CPX in water systems, similar to the tooth-backlash-inspired triboelectric nanogenerator (TB-TENG), which demonstrates that compact, self-powered sensors can achieve high diagnostic accuracy (98.4%) comparable to conventional methods while maintaining strong practical applicability [51].

4. Conclusions

Herein, we describe the successful synthesis of CD-AgNPs composite using Thai basil leaves-derived CDs and applied them for highly selective voltammetric detection of CPX. Structural and elemental characterization confirmed the formation of uniformly distributed, spherical AgNPs exhibiting a mean particle size of 20.24 ± 2.5 nm. When these CD-AgNPs were immobilized on a GCE, the electroactive surface area increased by 54.02% relative to the bare electrode, the CD-AgNPs/GCE sensor exhibited a 42.1% rise in anodic peak current than bare GCE. Electrochemical analysis revealed that the oxidation process of CPX follows a diffusion-controlled mechanism involving two electrons and two protons, with a k^0 of 0.1157 s^{-1} and a E^0 of 1.077 V. The Γ^* of CPX on the CD-AgNPs/GCE was calculated to be $6.593 \times 10^{-11} \text{ mol cm}^{-2}$. The developed sensor showed superior detection capability of 0.1124 μM (LOQ = 0.3746 μM ; sensitivity = $0.7977 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$) and 0.3237 μM (LOQ = 1.079 μM ; sensitivity = $0.2770 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$) within the respective concentration intervals spanning 0.25–40 μM and 40–200 μM . The CD-AgNPs/GCE also exhibited remarkable discrimination toward CPX against potential interfering analytes such as phenylbutazone, chloroquine, amoxicillin, chloramphenicol, and sulfamethoxazole. Stability assessments revealed only a 11.4% decrease in peak current response after 10 repeated cycles. When validated against HPLC, the deviation in detected CPX concentrations in water samples collected from the river and tap was limited to 6.4% and 8.1%, respectively, with

Table 2

Evaluation of CD-AgNPs/GCE performance for CPX detection in water samples from tap and river sources, besides a comparative analysis employing HPLC and calculation of Student's *t*-Test.

Sample	CPX spiked (μM)	Performance of CD-AgNPs/GCE			Determination of CPX using HPLC (μM)	Deviation (%)	<i>t</i> -value
		Determination of CPX (μM) (*n)	Recovery (%)	RSD (%)			
Tap water	5	5.29	105.8	5.2	4.95	6.4	1.83
	20	19.84	99.2	4.9	20.56	−3.6	0.29
	40	40.83	102.1	3.5	38.85	4.8	1.01
	80	79.88	99.9	4.4	81.12	−1.6	0.06
River water	5	5.32	106.4	6.8	4.89	8.1	1.53
	20	20.68	103.4	5.3	19.85	4.0	1.07
	40	41.92	104.8	3.8	39.78	5.1	2.09
	80	81.73	102.2	4.6	79.73	2.4	0.80

* n: Average of three measured samples.

near 100% recovery in both matrices. These findings underscore the potential of the CD-AgNPs/GCE platform as a robust, field-deployable sensor for detecting pharmaceutical residues in environmental samples.

CRedit authorship contribution statement

Swagata Patra: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Seokwoo Jeon:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization. **Weraut Srituravanich:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jwpe.2026.110030>.

Data availability

Data will be made available on request.

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